

CCT College Dublin Continuous Assessment

Programme Title:	L7 Dip. in Data Analytics - Sept 2025 cohort		
Delivery Mode:	PT		
Cohort Details:	<i>MachLearn_L7DipData_Sept25, Term 2</i>		
Module Title(s):	<i>Machine Learning</i> <i>Programme schedules are all published on the CCT IQR Provider Profile</i>		
Assignment Type:	<i>Individual</i>	Weighting(s):	<i>60%</i>
Assignment Title:	<i>CA2</i>		
Lecturer(s):	<i>Dr. Muhammad Iqbal</i>		
Issue Date:	<i>Monday 23rd February 2026</i>		
Submission Deadline Date:	<i>Friday 3rd April 2026@11:59</i>		
Late Submission Penalty:	Late submissions will be accepted up to 5 calendar days after the deadline. All late submissions are subject to a penalty of 10% of the mark awarded. Submissions received more than 5 calendar days after the deadline above <u>will not</u> be accepted and a mark of 0% will be awarded.		
Method of Submission:	This assignment is submitted via Moodle.		
Instructions for Submission:	<i>Upload all files separately, such as MS word file, jupyter notebook and any supporting information on Moodle.</i>		
Feedback Method:	Results posted in Moodle gradebook		
Feedback Date:	<i>After approval from Exams board.</i>		

Assessment Outline

The GitHub Classroom link for CA2 is provided below and should be used for development and version control purposes only:

<https://classroom.github.com/a/lprccbID>

Students must upload all required files to Moodle for CA2 assessment, excluding the datasets. Any file that is present on GitHub but not uploaded to Moodle will not be considered for grading.

Assessment Task

This is an individual assessment focused on the practical implementation of machine learning models and natural language processing using the Python programming language. All programming work must be implemented in Python (Jupyter Notebook in Anaconda environment), with an emphasis on model optimisation, evaluation, and interpretation.

Task 1: Supervised Learning on the Abalone Dataset

- Using the Abalone dataset provided in CA1, implement a Neural Network and at least one additional machine learning model of your choice (e.g., Support Vector Machine, Random Forest, Ridge Regression, or Logistic Regression) for either a classification task or a regression task, as appropriate to the selected models using a split of 75% for training and 25% for testing.
- Explain the purpose of hyperparameter tuning. Apply GridSearchCV to tune hyperparameters for each chosen model by selecting at least two hyperparameters, with 2 - 3 candidate values for each, and identify the optimal configuration. Evaluate, compare, and interpret their performance for the chosen models using appropriate metrics. Your interpretation should highlight performance differences and discuss the strengths and limitations of each model.

(0 – 40 marks)

Task 2: Sentiment Analysis with NLP

A dataset of reviews or tweets (for example, yelp.csv) is provided on Moodle. Clean and preprocess the text data (e.g., remove punctuation, stopwords, URLs, and unnecessary characters) and then perform sentiment analysis using either the VADER or TextBlob library. Evaluate and interpret the resulting sentiment scores.

- Build a reusable NLP preprocessing pipeline including tokenization, stop-word removal, and lemmatization. Show before-and-after examples of cleaned text, compute and visualise basic text statistics.
- Transform the text data into numerical features using either Bag of Words or TF-IDF. Apply a suitable classification machine learning model to predict the sentiment of reviews or tweets and evaluate its performance using a classification report.

(0 – 40 marks)

Task 3: LLM Applications in Business/ Banking

Using publicly available online business or banking text data (for example, a book (pg32027.txt) or text corpus), apply a local open-source Large Language Model (LLM) to generate and answer five business- or banking-related questions based on the text. Briefly explain how this task demonstrates the role of LLMs in generative AI applications and identify three ethical issues associated with LLM usage along with the mitigation strategies applied in your implementation.

(0 – 20 marks)

Assessment Requirements

All assessment submissions must meet the following minimum requirements:

- All files (MS word, Jupyter notebook and any supporting document) should be uploaded separately on Moodle.
- Number of Words in the report (1500 words +/-10%) excluding diagrams, code, references, citations, Table of Contents (TOCs) and titles.
- Clearly detail the number of words used in the report.
- You must use the classroom GitHub link provided for version control. You should have at least 5 commits on GitHub before submission.
- The rubric is provided for the detailed breakdown of marks at the end of this CA2.
- Use [Harvard Referencing](#) when citing third party material
- Be the student's own work.
- Include the CCT assessment cover page.
- Be submitted by the deadline date specified or be subject to late submission penalties.

Learning Outcomes:

This assessment addresses the following module learning outcomes for this module:

- **MLO 4:** Optimize the performance of classification and regression models and fine-tune to ensure robustness in real-world applications.
- **MLO 5:** Demonstrate a proficiency in utilizing Natural Language Processing (NLP) for the analysis and extraction of insights from textual data.
- **MLO 6:** Describe the significance of Large Language Models (LLMs) in generative AI applications and apply them to selected scenarios while adhering to ethical guidelines.

Use of Artificial Intelligence

Artificial Intelligence (AI) can be used in this assignment in the following way(s):

- Brainstorming and refining your ideas;
- Fine tuning your research questions;
- Finding information on your topic;
- Drafting an outline to organise your thoughts;
- Checking grammar and style
- Other

Note for students:

- You are required to include all AI prompt(s) and outputs in an appendix, including a short reflection (approx. 150 words) on how AI enhanced your work, or otherwise. These words are not part of 1500 words asked for CA2.
- You are responsible for the information you submit based on an AI query. Your use of AI tools must be properly documented and cited.
- Any assignment that is found to have used generative AI tools in an unauthorised way will be subject to college disciplinary procedures as outlined in the QA Manual.
- Cite all AI references used in your assignment.
- You must declare how you used AI on your assignment cover sheet.
- In the case of any suspicious work submitted by a student, the student will be called for a Q&A session.
- When in doubt about permitted usage, please ask for clarification to your lecturer.

Grading Criteria

This grading rubric sets out the marking criteria for your assignment.

Assessment Rubric

Performance Level	Task 1: Supervised Learning on Abalone Dataset	Task 2: Sentiment Analysis with NLP	Task 3: LLM Applications in Business/Banking
Weighting	40%	40%	20%
Excellent (≥70%)	Neural Network and an additional model are correctly implemented for the selected task (regression or classification). At least two hyperparameters with 2 - 3 candidate values for each per model are systematically tuned using GridSearchCV. The purpose and importance of hyperparameter tuning are clearly explained. Models are evaluated using appropriate metrics, with strong comparison, critical interpretation, and clear discussion of strengths, limitations, and generalisation performance.	Text data is thoroughly cleaned using a well-designed, reusable NLP preprocessing pipeline (tokenisation, stop-word removal, lemmatisation). Clear before-and-after examples are shown. Text statistics are computed and effectively visualised. Sentiment analysis is correctly performed using VADER or TextBlob. Text is transformed using BoW or TF-IDF, and a suitable classification model is implemented and evaluated using a classification report with insightful interpretation.	A local open-source LLM is effectively applied to generate and answer five relevant business/banking questions grounded in the provided text. Responses clearly demonstrate evidence-based generation. The role of LLMs in generative AI is clearly explained. Three ethical issues are accurately identified, and mitigation strategies are explicitly implemented and justified within the system.
Very Good (60–69%)	Models are correctly implemented and tuned using GridSearchCV with multiple hyperparameters and candidate values. Explanation of tuning purpose is clear but requires a depth. Evaluation metrics are appropriate and results are well presented. Comparison between models is sound, though discussion of limitations and generalisation could be deeper.	NLP preprocessing pipeline is clearly implemented with before-and-after examples and text statistics. Sentiment analysis is correctly applied. Feature extraction and classification are evaluated appropriately, though interpretation and justification of preprocessing choices require more depth.	Local LLM application is clearly demonstrated with relevant generated questions and answers grounded in the text. The role of LLMs is explained clearly. Ethical issues and mitigation strategies are identified, though explanations and implementation links could be more detailed.
Good (50–59%)	Models are implemented with acceptable configuration and basic hyperparameter tuning including candidate values. Evaluation metrics are correctly reported, but comparison and interpretation require more analytical depth. Discussion of strengths and limitations is mostly descriptive.	A basic NLP pipeline is implemented. Sentiment analysis is performed correctly. Cleaned text examples and some text statistics are shown. Feature extraction and classification are correctly applied, but explanations and interpretation are limited.	LLM is applied at a basic level to generate and answer questions. Responses are generally relevant but weakly grounded in the text. Ethical issues are identified but discussion and mitigation strategies require more depth or implementation clarity.
Acceptable (40–49%)	Models are partially implemented with limited or ineffective hyperparameter tuning. Explanation of tuning is weak or unclear. Evaluation is performed but lacks appropriate metric selection or meaningful interpretation. Comparison between models is minimal.	Sentiment analysis is attempted with limited preprocessing. NLP pipeline is incomplete or poorly structured. Feature extraction and evaluation are present but weakly justified. Interpretation of results is superficial.	LLM usage is minimal or partially correct. Generated questions and answers are weakly linked to the source text. Ethical issues are mentioned but mitigation strategies are vague or not clearly connected to implementation.
Fail (<39%)	Models are incorrectly implemented or missing. Hyperparameter tuning is absent or incorrect. Evaluation and interpretation are missing or fundamentally flawed.	Sentiment analysis is incomplete or incorrect. NLP preprocessing, feature extraction, and evaluation are missing or improperly applied.	LLM task is incomplete, incorrect, or missing. Responses are not grounded in text. Ethical issues and mitigation strategies are not meaningfully addressed.

The Irish Grading System

The grading system in CCT is the QQI percentage grading system and is in common use in higher education institutions in Ireland. The pass mark and thresholds for different grade bands may be different from what you have experienced in the higher education system in other countries. CCT grades must be considered in the context of the grading system in Irish higher education and not assumed to represent the same standard the percentage grade reflects when awarded in an international context.

Additional Information

- Lecturers are not required to review draft assessment submissions. This may be offered at the lecturer's discretion.
- In accordance with CCT policy, feedback to learners may be provided in written, audio or video format and can be provided as individual learner feedback, small group feedback or whole class feedback.
- Results and feedback will only be issued when assessments have been marked and moderated / reviewed by a second examiner.
- Additional feedback may be provided as individual, small group or whole class feedback. Lecturers are not obliged to respond to email requests for additional feedback where this is not the specified process or to respond to further requests for feedback following the additional feedback.
- Following receipt of feedback, where a student believes there has been an error in the marks or feedback received, they should avail of the recheck and review process and should not attempt to get a revised mark / feedback by directly approaching the lecturer. Lecturers are not authorised to amend published marks outside of the recheck and review process or the Board of Examiners process.
- Students are advised that disagreement with an academic judgement is not grounds for review.
- For additional support with academic writing and referencing students are advised to contact the CCT Library Service.
- For additional support with subject matter content students are advised to contact the [CCT Student Mentoring Academy](#)
- For additional support with IT subject content, students are advised to access the [CCT Support Hub](#).

Assessment Cover Page

Student Full Name Lucas de Natale Pires

Student Number SBA25076

Module Title Machine Learning

Assessment Title CA2 - 60%

Lecturer/Supervisor Dr. Muhammad Iqbal

Assessment Due Date 04/04/2026

Date of Submission 04/04/2026

Use of AI Tools

I acknowledge the use of Claude for the purpose of spelling and grammar correction, reviewing longer texts for fluency and coherence, and translating content from Portuguese to English. The AI was explicitly instructed not to add any new content, only to identify what was unclear or incorrect."

Declaration

By submitting this assessment, I confirm that I have read the CCT policy on academic misconduct and understand the implications of submitting work that is not my own or does not appropriately reference material taken from a third party or other source.

I declare it to be my own work and that all material from third parties has been appropriately referenced.

I further confirm that this work has not previously been submitted for assessment by myself or someone else in CCT College Dublin or any other higher education institution.